Background Information

The purpose of this research project, "How do various clinical factors such as age, anemia, high blood pressure, CPK levels, diabetes, ejection fraction, platelet count, sex, serum creatinine and sodium levels, smoking status, and follow-up period correlate with mortality due to heart failure?" is critical for several reasons.

Global heart failure is concerning millions of individuals and maintains a high death rate. It made it to be a large public health concern.[1,3] Our research can be helpful in identifying high-risk patients, and, by examining the relationship between death and different clinical variables to provide guidance for therapy.

Previous studies pinpointed factors, such as age, sex, and geographical location and comorbidities as potential risk factors leading to higher mortality in heart failure patients.[1,2] However, our research is aiming to provide a more expanded analysis by studying a more extensive range of clinical variables, including those not previously studied extensively, like CPK levels, platelet count and serum sodium levels.

Incorporating unnecessary features and variables while evaluating heart failure risks may also exhibit unpredictable behavior. Conversely, it may reveal new risk profiles and interactions among factors, leading to deeper comprehension of how the disease progresses and the proposal of customized treatment plans.[4,5]

Existing Studies

Yes, there are several existing studies that have investigated the correlation between various clinical factors and mortality in heart failure patients, as evidenced by the research articles shared.

The study by Lee et al. developed and validated a predictive model for 30-day and 1-year mortality among hospitalized heart failure patients using clinical factors available at hospital presentation.[2] Multivariable predictors included older age, lower systolic blood pressure, higher respiratory rate, hyponatremia, and increased blood urea nitrogen levels.

The large global study by Johansson et al. examined the association between health-related quality of life (measured by the KCCQ-12 score) and mortality among 23,291 heart failure patients from 40 countries.[4] Lower KCCQ-12 scores, indicating poorer quality of life, were associated with higher risks of death and heart failure hospitalization across geographic regions.

Additionally, the systematic review by Emmons-Bell et al. identified 125 studies from 51 countries reporting prevalence, incidence, and/or mortality rates for heart failure published between 1990 and 2020.[3] While the primary focus was summarizing the epidemiological data, several included studies explored clinical and demographic factors associated with mortality in heart failure patients.

Bytyçi and Bajraktari provided an overview of mortality in heart failure patients, highlighting that despite significant improvements in treatment and management, mortality remains unacceptably high. Their review summarizes the evidence available up to 2015, noting that mortality is particularly high among elderly, male, and African American patients.[6]

Dataset

We plan to use the Heart Failure Clinical Records Dataset from the Faisalabad Institute of Cardiology and the Allied Hospital in Faisalabad, Pakistan. This dataset contains medical records of 299 heart failure patients collected during April–December 2015.[7,8]. The patients consisted of 105 women and 194 men, and their ages range between 40 and 95 years old. All 299 patients had left ventricular systolic dysfunction and had previous heart failures that put them in classes III or IV of New York Heart Association (NYHA) classification of the stages of heart failure.

Dependent Variables

The dependent variable is "death event", which is a binary variable indicating whether the patient died (1) or survived (0) before the end of the 130-day average follow-up period.[7,8]

Independent Variables

The independent variables are Age, Anemia (binary), High blood pressure (binary), Creatinine phosphokinase (CPK) levels , Diabetes (binary), Ejection fraction, Platelet count, Sex (binary), Serum creatinine, Serum sodium, Smoking status (binary), Follow-up period (time).

These factors were chosen based on their clinical importance and well-established connections with heart failure prognosis and mortality risk, which were validated by previous research studies in the area. [2,3,4] We discovered evidence that age, anemia, high blood pressure, CPK levels, diabetes, ejection fraction, platelet count, sex, serum creatinine, serum sodium, smoking status, and follow-up duration are associated with mortality outcomes in heart failure patients. Our choice of variables was reinforced by the Heart Failure Clinical Records Dataset's extensive set of demographics, clinical, and laboratory data, which exactly aligned with the topic of our study question.

Significance

Understanding the correlation between various clinical factors and mortality in heart failure patients is significant for improving risk stratification, treatment planning, and clinical decision-making. Identifying key risk factors can inform the development of predictive models, guide resource allocation, and contribute to the optimization of heart failure care and patient outcomes.[2-4]

Practical Implications

The practical implications of this research include Informing clinical practice guidelines and protocols for heart failure management by emphasizing monitoring and addressing identified risk factors.[2-4] Guiding the development of targeted preventive strategies and interventions aimed at modifiable risk factors. [2-4] Facilitating efficient resource allocation within healthcare systems through targeted screening and early intervention for high-risk individuals. [2-4] Potentially improving patient counseling and shared decision-making by providing quantitative estimates of prognosis based on individual risk profiles. [2-4]

Methodology

Descriptive Statistics

Before starting the actual descriptive statistics, we have checked if there are any missing or invalid characters in the dataset and found that there are no missing values in the dataset.

For descriptive statistics, we calculated the mean, standard deviation, median, and interquartile range for continuous variables, and frequencies and percentages for categorical variables in the heart failure dataset.

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
Age	40.00	51.00	60.00	60.83	70.00	95.00
Anemia	0	0	0	0.4314	1	1
Creatinine Phosphokinase	23.0	116.5	250.0	581.8	582.0	7861.0
Diabetes	0	0	0	0.4181	1	1
Ejection Fraction	14.00	30.00	38.00	38.08	45.00	80.00
High Blood Pressure	0	0	0	0.3512	1	1
Platelets	25100	212500	262000	263358	303500	850000
Serum Creatinine	0.500	0.900	1.100	1.394	1.400	9.400
Serum Sodium	113.0	134.0	137.0	136.6	140.0	148.0
Sex	0	0	1	0.6488	1	1
Smoking	0	0	0	0.3211	1	1
Time	4.0	73.0	115.0	130.3	203.0	285.0
DEATH_EVENT	0	0	0	0.3211	1	1

Table 1: Description of the dataset

We found that the median follow-up time was 115 days, with 67.56% of patients surviving the average 130-day follow-up period. The mean age was 60.83 years, with a median of 60 years, indicating a relatively symmetric distribution.

The distributions of CPK levels, ejection fraction, platelet count, serum creatinine, and follow-up time were right-skewed, while age and serum sodium were relatively symmetric. We also observed differences in mortality rates based on factors like high blood pressure and anemia from the frequency tables.

Data Visualization

We have first created distribution plots for continuous features “age", "creatinine phosphokinase", "ejection fraction", "platelets", "serum creatinine", "serum sodium" and "time".

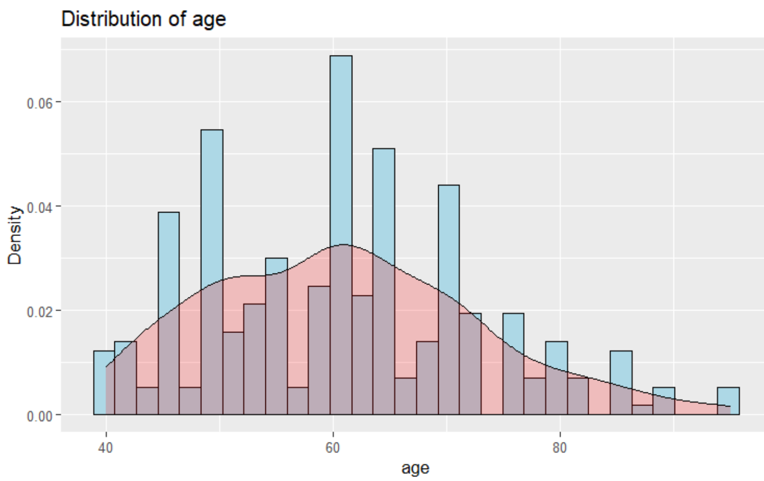

Figure 1: Distribution of age

The patient's ages range from 40 to 95. The average age is 60.83 years, with a median age of 60 years, indicating a rather symmetric distribution. The first and third quartiles are 51 and 70 years old, implying that roughly half of the patients are between those ages.

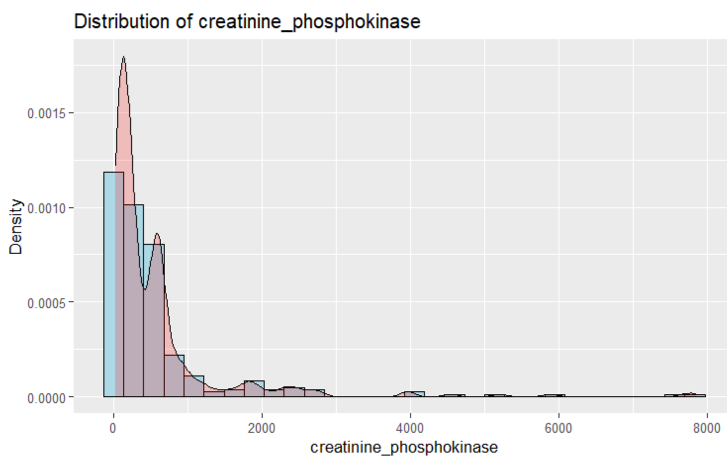

Figure 2: Distribution of Creatine Phosphokinase

CPK values range from 23 and 7861, with a median of 250 and a mean of 581.8. The distribution looks to be right-skewed, with the third quartile (582), which is closer to the highest value than the median.

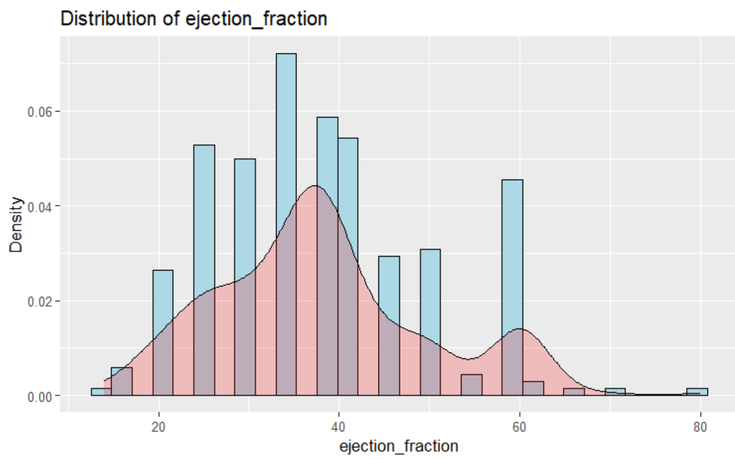

Figure 3: Distribution of Ejection Fraction

The ejection fraction ranges between 14% and 80%. The mean ejection percent is 38.08%, with a median of 38%, indicating a rather symmetric distribution. The first and third quartiles are 30% and 45%, respectively, indicating that roughly half of patients have an ejection percentage of 30% to 45%.

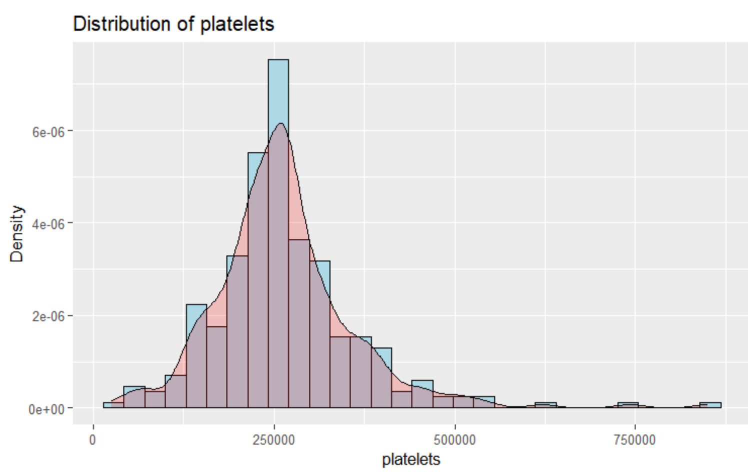

Figure 4: Distribution of Platelets

Platelet counts range from 25,100 and 850,000, with a median of 262,000 and a mean of 263,358. The distribution looks to be slightly right skewed, with the third quartile (303,500) being farther from the median than the first quartile (212,500).

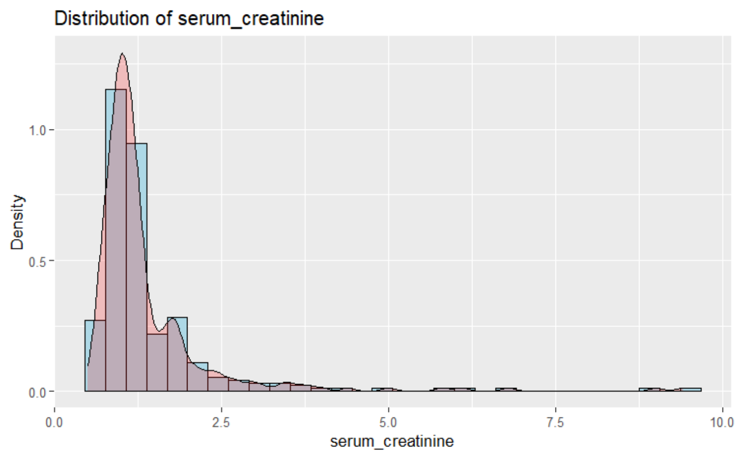

Figure 5: Distribution of Serum Creatine

Serum creatinine levels vary between 0.5 and 9.4, with a median of 1.1 and a mean of 1.394. The distribution appears to be right-skewed, with the third quartile (1.4) closer to the maximum value than to the median.

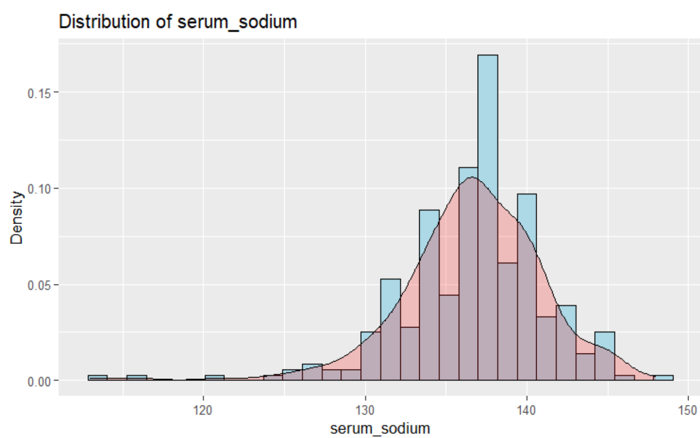

Figure 6: Distribution of Serum Sodium

The serum sodium level ranges from 113 to 148, with a median of 137 and a mean of 136.6. The distribution appears to be reasonably symmetric, with the mean and median being close together.

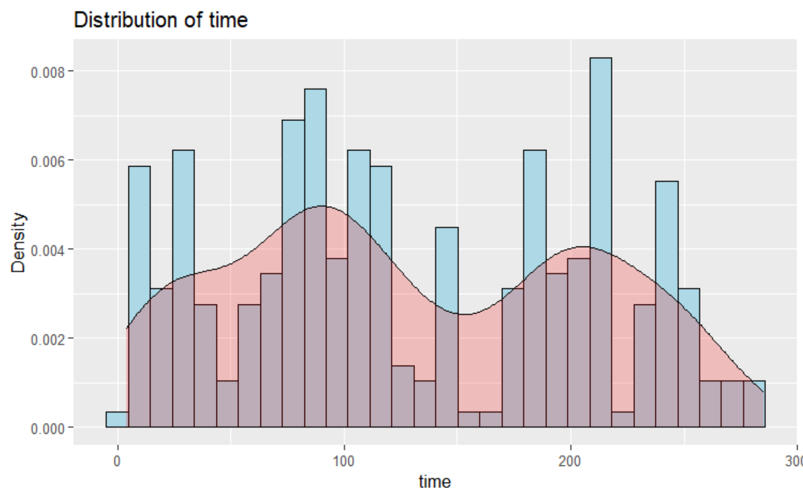

Figure 7: Distribution of Time

The follow-up time ranged between 4 and 285 days, with a median of 115 days and a mean of 130.3. The distribution looks to be right-skewed, with the third quartile (203 days) being farther from the median than the first quartile (73 days).

We created boxplots for continuous variables, revealing potential outliers for CPK levels, ejection fraction, platelet count, serum creatinine, and serum sodium.

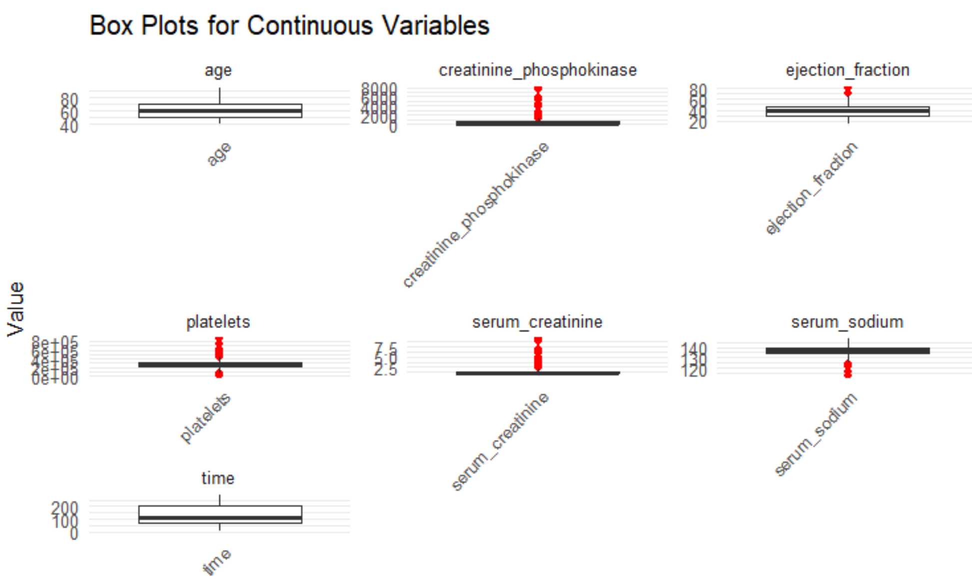

Figure 8: Boxplot of the numerical variables

The boxplots showed several very high CPK level values, two patients with unusually high ejection fraction values, several patients with extremely high platelet counts, and a few patients with very high serum creatinine levels, which were marked as outliers.

We also generated bar plots for categorical variables like anemia, diabetes, high blood pressure, sex, and smoking status, showing differences in mortality rates across the levels of these variables.

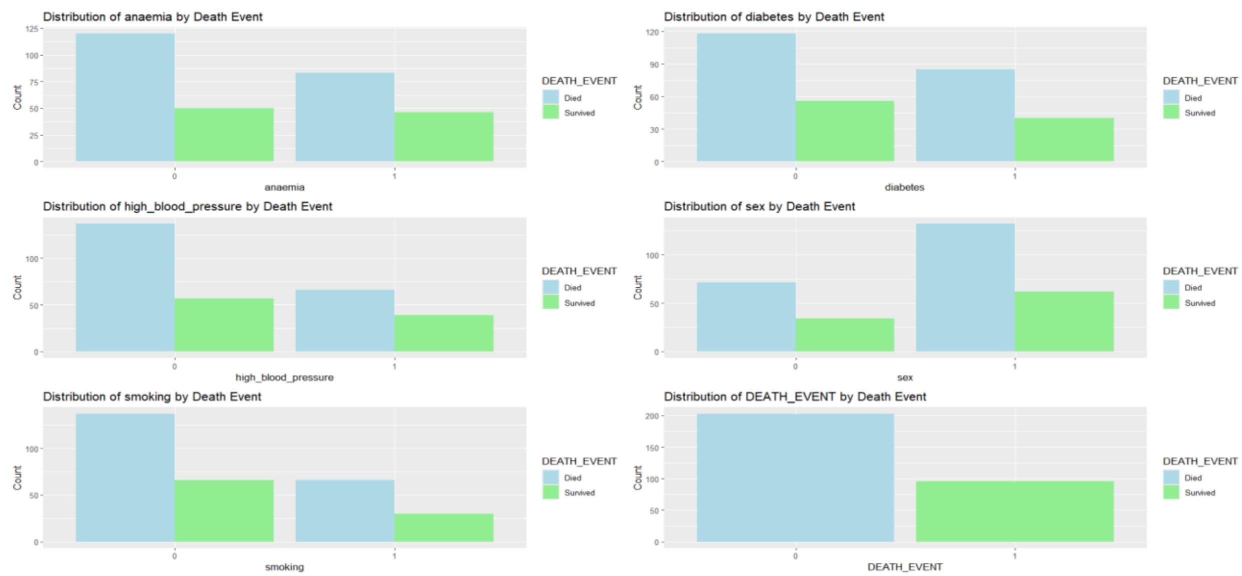

Figure 9: Bar plot of the categorical variables vs Death Event

For instance, 70.59% of patients without anemia survived, compared to 64.34% of patients with anemia, indicating that anemia may be associated with increased mortality risk. Visualizing these variables was interesting and relevant to the study, as they provided initial insights into potential relationships with the dependent variable (mortality).

To explore the potential relationship between continuous variables and the binary outcome (death event), we have created scatter plots with the continuous variable on the y-axis and the death event (0 or 1) on the x-axis. This can provide a visual representation of how the values of the continuous variable may differ between those who died and those who survived.

Figure 10: Distribution of Continuous Variables across Death Event.

When comparing time distribution with respect to death event, the death patients have tendency to have time below 0 and survival patients has higher time when compared. Time can be expected as a predictor.

Data Management

As a first step in data management, we have decided to first identify outliers using standard residuals for continuous variables, such as creatinine phosphokinase (CPK) levels, ejection fraction, platelet count, and serum creatinine. Box plots were used to visualize the distribution of these variables and detect extreme values that could potentially skew the analysis.

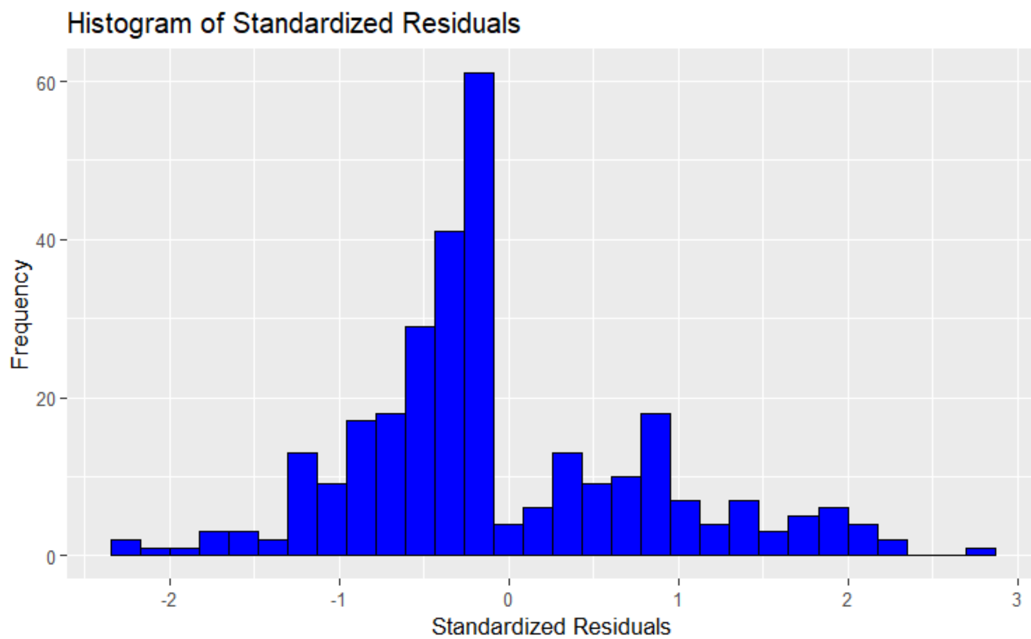

Figure 11: Distribution of Standardized Residuals.

Additionally, the standardized residuals from a logistic regression model were calculated, and observations with standardized residuals greater than 1.96 in absolute value were filtered out. This step helped remove 11 potential outliers, including cases with very high CPK levels, severe kidney dysfunction, low ejection fractions, electrolyte imbalances, and older patients with multiple comorbidities.

Next, the continuous variables (age, CPK levels, ejection fraction, platelet count, serum creatinine, serum sodium, and follow-up time) were standardized by subtracting the mean and dividing by the standard deviation.

```
       age.V1          creatinine_phosphokinase.V1 ejection_fraction.V1    platelets.V1      serum_creatinine.V1   serum_sodium.V1
 Min.   :-1.7405685   Min.   :-0.560557            Min.   :-2.047564    Min.   :-2.427675   Min.   :-0.865644   Min.   :-5.378043
 1st Qu.:-0.8237387   1st Qu.:-0.465637            1st Qu.:-0.692909    1st Qu.:-0.522450   1st Qu.:-0.470767   1st Qu.:-0.603875
 Median :-0.0736051   Median :-0.329447            Median :-0.015581    Median :-0.017334   Median :-0.273329   Median : 0.078149
 Mean   : 0.0000000   Mean   : 0.000000            Mean   : 0.000000    Mean   : 0.000000   Mean   : 0.000000   Mean   : 0.000000
 3rd Qu.: 0.7598766   3rd Qu.: 0.016185            3rd Qu.: 0.577081    3rd Qu.: 0.396404   3rd Qu.: 0.022829   3rd Qu.: 0.760172
 Max.   : 2.8435809   Max.   : 7.526212            Max.   : 3.540390    Max.   : 5.947599   Max.   : 7.920369   Max.   : 2.578903
      time.V1
 Min.   :-1.6179612
 1st Qu.:-0.7328034
 Median :-0.2145837
 Mean   : 0.0000000
 3rd Qu.: 0.9312933
 Max.   : 1.9999202
```

Figure 12: Data Description after Standardization

Standardization ensures that the variables are on the same scale, making the coefficients in the regression model more interpretable and preventing potential issues with numerical instability during model fitting.

Statistical tests and methods

1. Chi-Square Test for Categorical Variables

Much existing research states that high blood pressure and anemia are major factors for heart diseases[10,11]. High blood pressure is one of the major risk factors for cardiovascular diseases, contributing significantly to the overall disease burden[10]. The study by Fuchs and Whelton highlights the causative role of elevated blood pressure in various conditions like heart failure, atrial fibrillation, and stroke. Separately, the review by Silverberg et al[11] discusses the interplay between anemia, congestive heart failure, and kidney disease, forming a vicious cycle termed the "cardio-renal anemia syndrome." Effective management of both hypertension and anemia through interventions like antihypertensives, erythropoietin, and intravenous iron has been shown to improve cardiac function, reduce hospitalizations, and potentially slow disease progression[11]. So, we have started by let's check the relation between high blood pressure, anemia, and death.

Variable	P Value	Result
Anemia	0.4042	No significant difference
High Blood Pressure	0.1077	No significant difference
Diabetes	1	No significant difference
Sex	0.9881	No significant difference
Smoking	1	No significant difference

Table 2: Chi Squared test results on categorical variables

The tests did not find a statistically significant link between anemia (p-value = 0.4042), high blood pressure (p-value = 0.1077), and diabetes (p-value = 1) and the death event in this dataset. This implies that these individual characteristics may not have a significant independent effect on death in this group of heart failure patients.

The test for diabetes has a p-value of 1, indicating a complete lack of connection between this variable and mortality under the conditions of this group.

2. Fisher's Test on Categorical Variables – Sex and Smoking

Instead of using the chi-square test for the categorical variables "sex" and "smoking," we used Fisher's exact test. Fisher's exact test is recommended when dealing with small sample sizes or when the predicted counts in the contingency table are low, because it determines the precise probability of witnessing the given data or more extreme deviations from the null hypothesis without using approximation.

Variable	P Value	Result	95% Confidence Interval for Odds Ratio
Sex	1	No significant difference	(0.5962372, 1.8377372)
Smoking	1	No significant difference	(0.5615575, 1.7523316)

Table 3: Fisher's test results on categorical variables

A p-value of 1 for both variables indicates that there is insufficient evidence to reject the null hypothesis of no association between these variables and the death event. Furthermore, the confidence intervals for the odds ratios contain 1, supporting the conclusion that there is no statistically significant relationship between sex, smoking status, and death in this sample of heart failure patients.

3. Logistic Regression Model Test for Continuous Variables

We tested the association of the continuous variables with the death event using a logistic regression model. We picked this method because logistic regression is appropriate for examining the association between one or more independent variables and a binary dependent variable, in this case death.

We evaluated the following continuous variables: age, ejection fraction, serum sodium, creatinine phosphokinase, platelets, serum creatinine, and time. The results of the logistic regression model are shown in Table 3.

Feature Name	P-value	Finding
age	< 0.001	Significant association
ejection_fraction	< 0.001	Significant association
serum_sodium	0.001	Significant association
creatinine_phosphokinase	0.334	No significant association
platelets	0.551	No significant association
serum_creatinine	< 0.001	Significant association
time	< 0.001	Significant association

Table 3: Regression test results on continuous variables

Our analysis revealed that age, ejection fraction, serum sodium, serum creatinine, and time had significant associations with the death event, as indicated by their p-values being less than 0.001 or 0.001. These findings suggest that these variables may play a crucial role in predicting the likelihood of death in heart failure patients.

On the other hand, creatinine phosphokinase and platelets did not show a significant association with the death event, as their p-values were greater than the commonly accepted threshold of 0.05.

These results are interesting and helpful for our predictive logistic model because they highlight the key continuous variables that should be included in the model to improve its predictive accuracy. By incorporating age, ejection fraction, serum sodium, serum creatinine, and time as predictors, we can potentially build a more robust and reliable model for predicting mortality in heart failure patients.

However, it's important to note that these findings are based solely on the statistical significance of the variables, and further analysis may be required to assess their individual contributions and potential interactions within the logistic regression model.

Logistic Model

To build the logistic regression model for predictive analysis, we first split the data into training and testing sets. We used 75% of the data for training and the remaining 25% for testing.

1. **Logistic Model with One Variable**

We started by building logistic regression models with each of the significant continuous variables individually. This helps us understand the predictive power of each variable. Table 4 shows the accuracy of the logistic regression model for each variable individually. This gives us an idea of the predictive power of each variable on its own in relation to mortality due to heart failure.

Variable	**Accuracy**
Age	0.7129630
Ejection fraction	0.7407407
Serum creatinine	0.7129630
Serum sodium	0.6898148

Time	0.8518519

Table 4: Test Accuracies of the respective models

The highest accuracy of 0.8518519 is achieved with the 'time' variable, suggesting that the follow-up period alone has relatively strong predictive power for mortality in heart failure patients. The second-highest accuracy of 0.7407407 is for the 'ejection_fraction' variable, indicating that ejection fraction alone is also a reasonably good predictor of mortality.

2. Logistic Model with Multiple Variables

Next, we built a logistic regression model using all the significant variables together. The summary of the model provides information about the coefficients and their significance.

Variable	**Estimate**	**Std. Error**	**z value**	**Pr(>|z|)**
Intercept	-1.9484	0.3257	-5.982	2.20e-09
age	0.5216	0. 2406	2.168	0.03019
ejection_fraction	-1.2453	0.2760	-4.512	6.42e-06
serum_creatinine	0.8945	0.2867	3.120	0.00181
serum_sodium	-0.4392	0.2227	-1.972	0.04856
time	-2.3273	0.3709	-6.275	3.49e-10

Table 5: Model summary of logistic model

We used the trained model to make predictions on the test data, obtaining the predicted probabilities and predicted classes (0 or 1).

The confusion matrix shows that the model correctly identified 52 true negatives (patients who did not die) and 12 true positives (patients who died). However, it also had 7 false negatives (patients who died but were predicted as not dying) and 1 false positive.

The model achieved an overall accuracy of 0.8889, which is reasonably good. However, in medical contexts, false negatives are more concerning as missing a real case could have severe consequences. The model's sensitivity (0.9811) is high, indicating a good ability to detect positive cases. However, the specificity (0.6316) is relatively lower, suggesting a higher rate of false positives.

The area under the ROC curve (AUC) of 0.9603 indicates excellent overall performance of the model in distinguishing between the positive (mortality) and negative (survival) classes.

Conclusions

Regarding the research question, "How do various clinical factors such as age, anemia, high blood pressure, CPK levels, diabetes, ejection fraction, platelet count, sex, serum creatinine and sodium levels, smoking status, and follow-up period correlate with mortality due to heart failure?", the results from these models provide valuable insights:

1. Age, ejection fraction, serum creatinine, serum sodium, and follow-up period (time) emerged as significant predictors of mortality, indicating a strong correlation between these factors and mortality in heart failure patients.
2. The individual variable accuracies suggest that follow-up period (time) and ejection fraction have the strongest predictive power for mortality when considered individually.
3. The negative coefficients for ejection fraction, serum sodium, and time in the multiple variable model indicate that lower values of these variables are associated with an increased risk of mortality.
4. The positive coefficients for age and serum creatinine suggest that higher values of these variables are associated with a higher risk of mortality.

These findings highlight the importance of considering these clinical factors, particularly age, ejection fraction, serum creatinine, serum sodium, and follow-up period, in assessing mortality risk and developing predictive models for heart failure patients.

References

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Appendix

Installing the Required Packages

```
{r}
# Installing the required packages
install.packages("ggplot2")  # For data visualization
install.packages("moments")  # For calculating skewness and kurtosis
```

```
Error in install.packages : Updating loaded packages
Error in install.packages : Updating loaded packages
```

```
{r}
install.packages("GGally")
```

```
Error in install.packages : Updating loaded packages
```

```
{r}
#loading the packages
library(tidyverse)
library(ggplot2)
library(moments)
library(GGally)
library(dplyr)
library(caret)
library(pROC)
```

Import the dataset

```
{r}
heart_failure_data <- read.csv("heart_failure_clinical_records_dataset.csv")
```

```
{r}
head(heart_failure_data)
```

Description: df [6 x 13]

Description: df [6 × 13]

	age <dbl>	anaemia <int>	creatinine_... <int>	diabetes <int>	ejection_fr... <int>
1	75	0	582	0	20
2	55	0	7861	0	38
3	65	0	146	0	20
4	50	1	111	0	20
5	65	1	160	1	20
6	90	1	47	0	40

6 rows | 1-6 of 13 columns

```
{r}
##Getting the data structure
str(heart_failure_data)
```

```
'data.frame':	299 obs. of  13 variables:
 $ age                     : num  75 55 65 50 65 90 75 60 65 80 ...
 $ anaemia                 : int  0 0 0 1 1 1 1 1 0 1 ...
 $ creatinine_phosphokinase: int  582 7861 146 111 160 47 246 315 157 123 ...
 $ diabetes                : int  0 0 0 0 1 0 0 1 0 0 ...
 $ ejection_fraction       : int  20 38 20 20 20 40 15 60 65 35 ...
 $ high_blood_pressure     : int  1 0 0 0 0 1 0 0 0 1 ...
 $ platelets               : num  265000 263358 162000 210000 327000 ...
 $ serum_creatinine        : num  1.9 1.1 1.3 1.9 2.7 2.1 1.2 1.1 1.5 9.4 ...
 $ serum_sodium            : int  130 136 129 137 116 132 137 131 138 133 ...
 $ sex                     : int  1 1 1 1 0 1 1 1 0 1 ...
 $ smoking                 : int  0 0 1 0 0 1 0 1 0 1 ...
 $ time                    : int  4 6 7 7 8 8 10 10 10 10 ...
 $ DEATH_EVENT             : int  1 1 1 1 1 1 1 1 1 1 ...
```

The dataset has 299 observations (rows) and 13 variables (columns).

All the existing categorical features are already encoded so no need of label encoding.

```
{r}
#summary of the dataset
summary(heart_failure_data)
```

```
      age           anaemia       creatinine_phosphokinase    diabetes     
ejection_fraction high_blood_pressure   platelets    
 Min.   :40.00   Min.   :0.0000   Min.   :  23.0           Min.   :0.0000   Min.   
:14.00     Min.   :0.0000       Min.   : 25100  
 1st Qu.:51.00   1st Qu.:0.0000   1st Qu.: 116.5           1st Qu.:0.0000   1st 
Qu.:30.00     1st Qu.:0.0000       1st Qu.:212500  
 Median :60.00   Median :0.0000   Median : 250.0           Median :0.0000   
Median :38.00     Median :0.0000       Median :262000  
 Mean   :60.83   Mean   :0.4314   Mean   : 581.8           Mean   :0.4181   Mean   
:38.08     Mean   :0.3512       Mean   :263358  
 3rd Qu.:70.00   3rd Qu.:1.0000   3rd Qu.: 582.0           3rd Qu.:1.0000   3rd 
Qu.:45.00     3rd Qu.:1.0000       3rd Qu.:303500  
 Max.   :95.00   Max.   :1.0000   Max.   :7861.0           Max.   :1.0000   Max.   
:80.00     Max.   :1.0000       Max.   :850000  
 serum_creatinine  serum_sodium       sex            smoking            time      
DEATH_EVENT    
 Min.   :0.500    Min.   :113.0   Min.   :0.0000   Min.   :0.0000   Min.   :  4.0  
Min.   :0.0000  
 1st Qu.:0.900    1st Qu.:134.0   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.: 73.0  
1st Qu.:0.0000  
 Median :1.100    Median :137.0   Median :1.0000   Median :0.0000   Median :115.0  
Median :0.0000  
 Mean   :1.394    Mean   :136.6   Mean   :0.6488   Mean   :0.3211   Mean   :130.3  
Mean   :0.3211  
 3rd Qu.:1.400    3rd Qu.:140.0   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:203.0  
3rd Qu.:1.0000  
 Max.   :9.400    Max.   :148.0   Max.   :1.0000   Max.   :1.0000   Max.   :285.0  
Max.   :1.0000  
```

```
{r}
#checking for null values
sum(is.na(heart_failure_data))
```

```
[1] 0
```

No missing values in the dataset.

Converting into factor for all categorical features

```
{r}
heart_failure_data$anaemia <- as.factor(heart_failure_data$anaemia)
heart_failure_data$diabetes <- as.factor(heart_failure_data$diabetes)
heart_failure_data$high_blood_pressure <- as.factor
(heart_failure_data$high_blood_pressure)
heart_failure_data$sex <- as.factor(heart_failure_data$sex)
heart_failure_data$smoking <- as.factor(heart_failure_data$smoking)
heart_failure_data$DEATH_EVENT <- as.factor(heart_failure_data$DEATH_EVENT)
```

```
{r}
#now check the structure
str(heart_failure_data)
```

```
'data.frame':	299 obs. of  13 variables:
 $ age                     : num  75 55 65 50 65 90 75 60 65 80 ...
 $ anaemia                 : Factor w/ 2 levels "0","1": 1 1 1 2 2 2 2 2 1 2 ...
 $ creatinine_phosphokinase: int  582 7861 146 111 160 47 246 315 157 123 ...
 $ diabetes                : Factor w/ 2 levels "0","1": 1 1 1 1 2 1 1 2 1 1 ...
 $ ejection_fraction       : int  20 38 20 20 20 40 15 60 65 35 ...
 $ high_blood_pressure     : Factor w/ 2 levels "0","1": 2 1 1 1 1 2 1 1 1 2 ...
 $ platelets               : num  265000 263358 162000 210000 327000 ...
 $ serum_creatinine        : num  1.9 1.1 1.3 1.9 2.7 2.1 1.2 1.1 1.5 9.4 ...
 $ serum_sodium            : int  130 136 129 137 116 132 137 131 138 133 ...
 $ sex                     : Factor w/ 2 levels "0","1": 2 2 2 2 1 2 2 2 1 2 ...
 $ smoking                 : Factor w/ 2 levels "0","1": 1 1 2 1 1 2 1 2 1 2 ...
 $ time                    : int  4 6 7 7 8 8 10 10 10 10 ...
 $ DEATH_EVENT             : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 2 2 2 2 ...
```

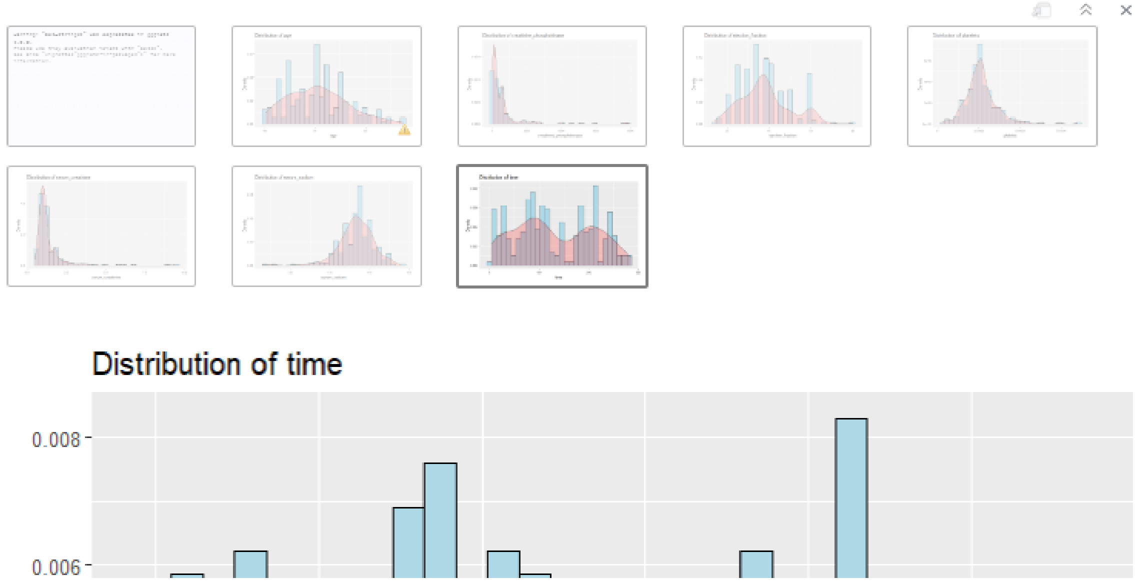
Distribution Plots

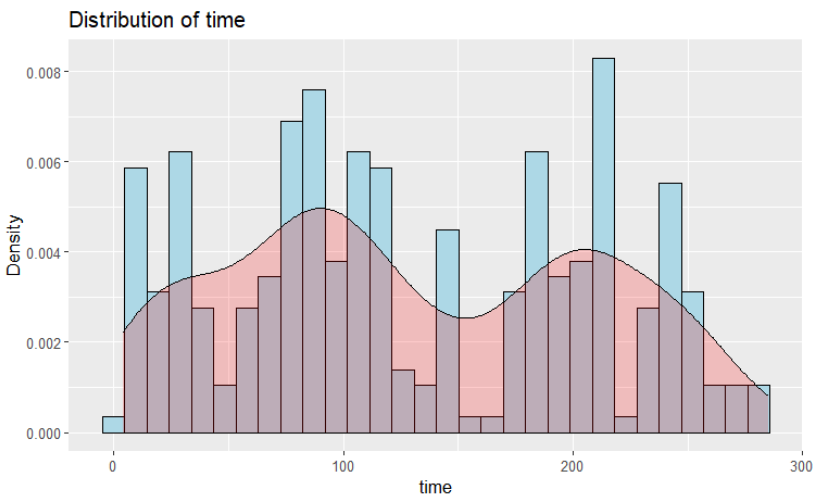
```
{r}

# defining the Continuous variables
continuous_vars <- c("age", "creatinine_phosphokinase", "ejection_fraction",
```

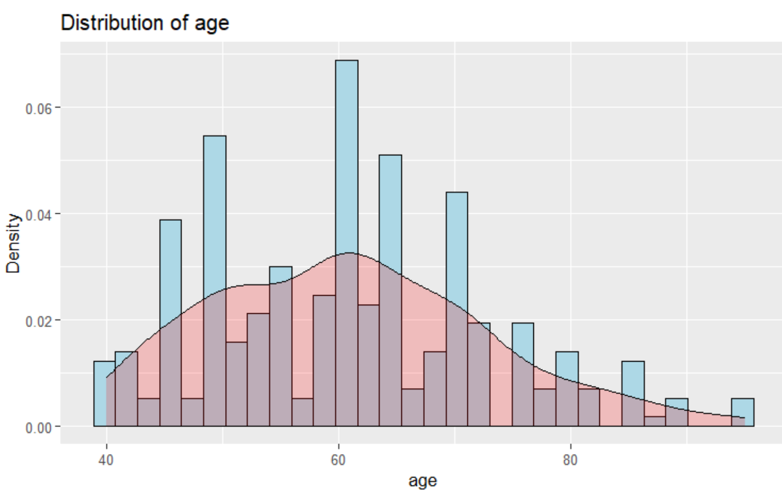
Distribution Plots

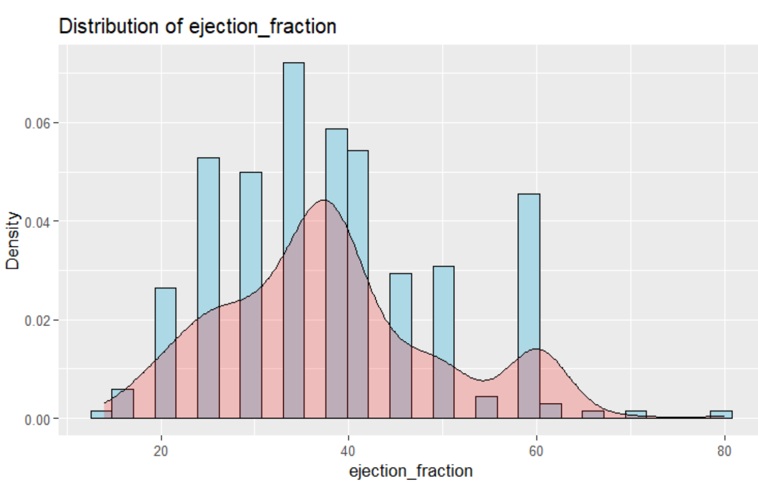
```
{r}

# defining the Continuous variables
continuous_vars <- c("age", "creatinine_phosphokinase", "ejection_fraction",
"platelets", "serum_creatinine", "serum_sodium", "time")
```

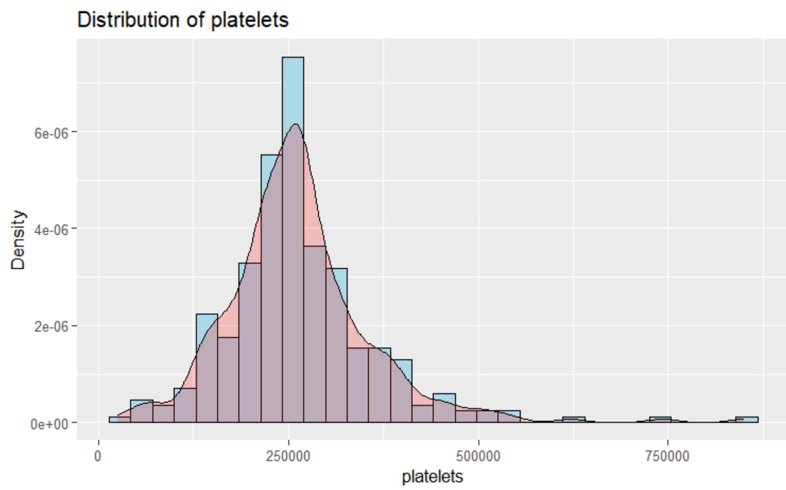
```
{r}
for (var in continuous_vars) {
  p <- ggplot(heart_failure_data, aes_string(x = var)) +
    geom_histogram(aes(y = ..density..), fill = "lightblue", color = "black", bins
= 30) +
    geom_density(alpha = 0.2, fill = "red") +
    ggtitle(paste("Distribution of", var)) +
    xlab(var) +
    ylab("Density")
  print(p)
}
```

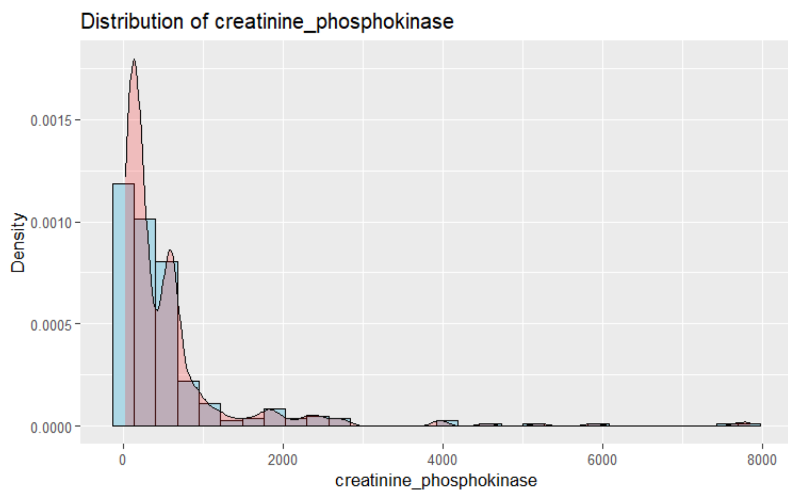


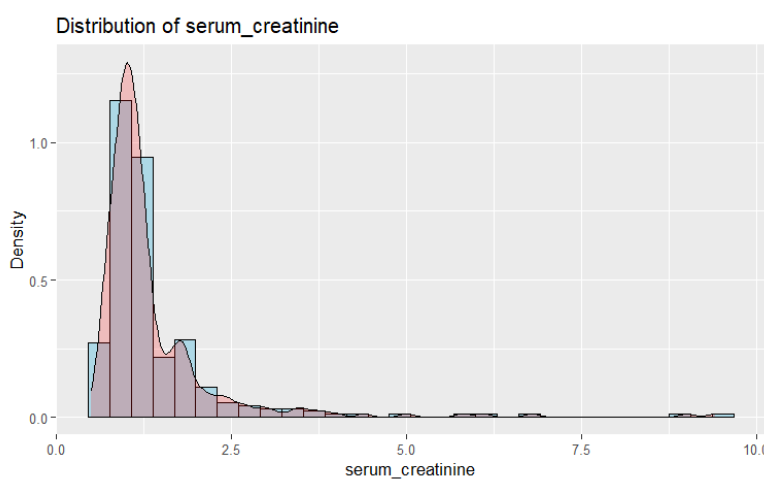

Distribution of time
Density
time
0.000
0.002
0.004
0.006
0.008
0
100
200
300

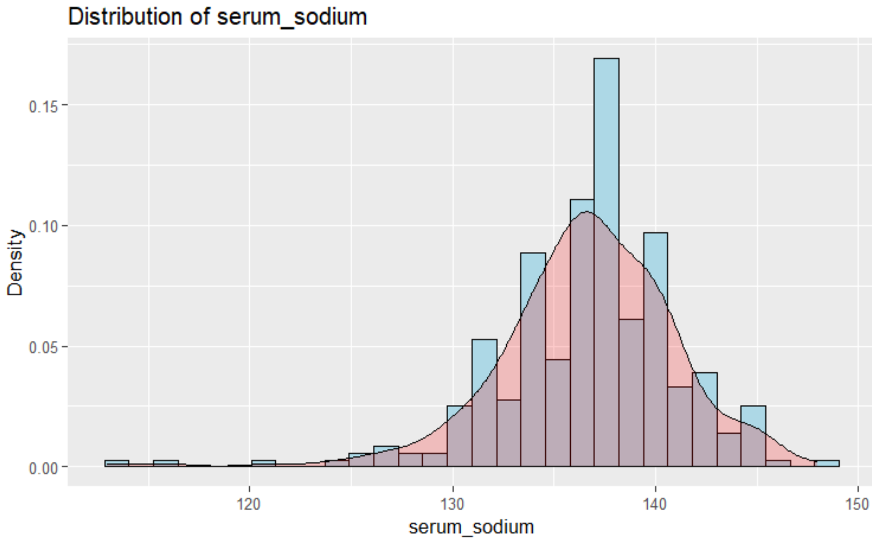

Distribution of age
Density
age
0.00
0.02
0.04
0.06
40
60
80


Distribution of ejection_fraction
Density
ejection_fraction
0.00
0.02
0.04
0.06
20
40
60
80


Distribution of platelets
Density
6e-06
4e-06
2e-06
0e+00
0
250000
500000
750000
platelets


Distribution of creatinine_phosphokinase
Density
0.0015
0.0010
0.0005
0.0000
0
2000
4000
6000
8000
creatinine_phosphokinase


Distribution of serum_creatinine
Density
1.0
0.5
0.0
0.0
2.5
5.0
7.5
10.0
serum_creatinine


Distribution of serum_sodium
Density
0.15
0.10
0.05
0.00
120
130
140
150
serum_sodium

Age: The patient's ages range from 40 to 95. The average age is 60.83 years, with a median age of 60 years, indicating a rather symmetric distribution. The first and third quartiles are 51 and 70 years old, implying that roughly half of the patients are between those ages.

Creatinine Phosphokinase (CPK) Levels: CPK values range from 23 and 7861, with a median of 250 and a mean of 581.8. The distribution looks to be right-skewed, with the third quartile (582), which is closer to the highest value than the median.

Ejection Fraction: The ejection fraction ranges between 14% and 80%. The mean ejection percent is 38.08%, with a median of 38%, indicating a rather symmetric distribution. The first and third quartiles are 30% and 45%, respectively, indicating that roughly half of patients have an ejection percentage of 30% to 45%.

Platelets: Platelet counts range from 25,100 and 850,000, with a median of 262,000 and a mean of 263,358. The distribution looks to be slightly right-skewed, with the third quartile (303,500) being farther from the median than the first quartile (212,500).

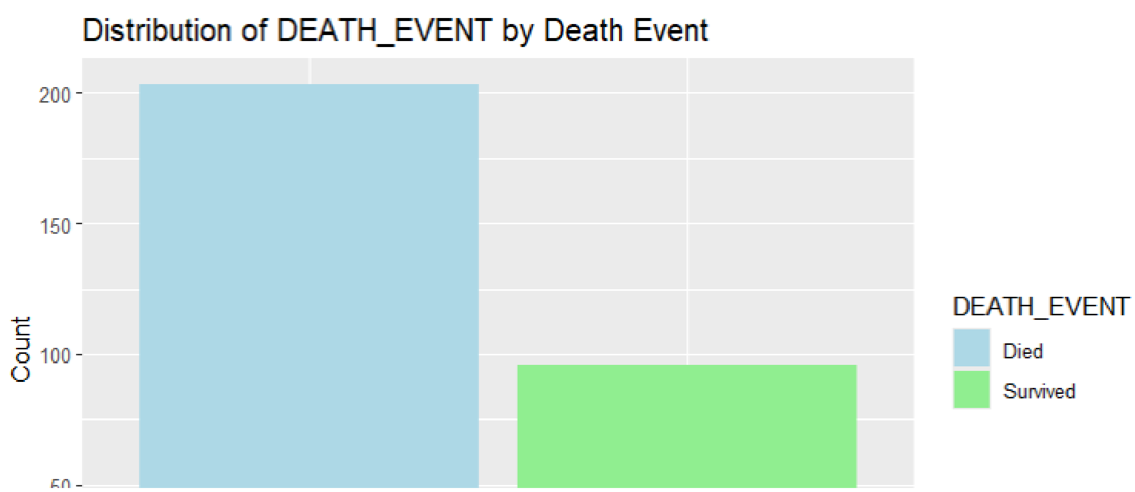
Serum Creatinine: Serum creatinine levels vary between 0.5 and 9.4, with a median of 1.1 and a mean of 1.394. The distribution appears to be right-skewed, with the third quartile (1.4) closer to the maximum value than to the median.

Serum Sodium: The serum sodium level ranges from 113 to 148, with a median of 137 and a mean of 136.6. The distribution appears to be reasonably symmetric, with the mean and median being close together.

Time: The follow-up time ranged between 4 and 285 days, with a median of 115 days and a mean of 130.3. The distribution looks to be right-skewed, with the third quartile (203 days) being farther from the median than the first quartile (73 days).

```
{r}
#defining the categorical features
categorical_vars <- c("anaemia", "diabetes", "high_blood_pressure", "sex",
"smoking", "DEATH_EVENT")
```

```{r}
for (var in categorical_vars) {
  p <- ggplot(heart_failure_data, aes_string(x = var, fill = "DEATH_EVENT")) +
    geom_bar(position = "dodge") +
    scale_fill_manual(values = c("lightblue", "lightgreen"), labels = c("Died", 
"Survived")) +
    ggtitle(paste("Distribution of", var, "by Death Event")) +
    xlab(var) +
    ylab("Count")
  print(p)
}
```



- **Anemia**: 70.59% of patients without anemia survived, compared to 64.34% of patients with anemia. Anemia appears to be connected with an increased death risk.
- **Diabetes**: The survival rates for patients without and with diabetes were comparable (67.82% and 68%, respectively), indicating that diabetes may not have a significant impact on mortality in this cohort.
- **High Blood Pressure**: Patients without high blood pressure had a greater survival rate (70.62%) than those with high blood pressure (62.86%), implying that high blood pressure is related with increased mortality.
- **Sex**: Males survived at a higher rate (68.04%) than females (67.62%), albeit the difference was modest.
- **Smoking**: The survival rate was somewhat higher for nonsmokers (67.49%) than smokers (68.75%), but the difference was not significant.
- **DEATH_EVENT**: The dataset is imbalanced. With Majority class as death.

Key Interpretation

High Blood Pressure and Anemia are highly effecting the survival of the patients.

Checking Correlations

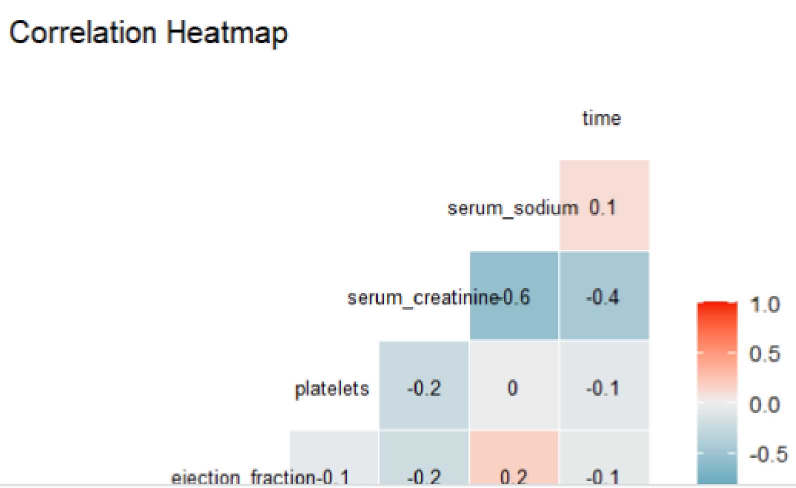
```
{r}
num_cols <- unlist(lapply(heart_failure_data, is.numeric))
heart_failure_data_num <- heart_failure_data[, num_cols]
```

```
{r}
cor_matrix <- cor(heart_failure_data_num)
```

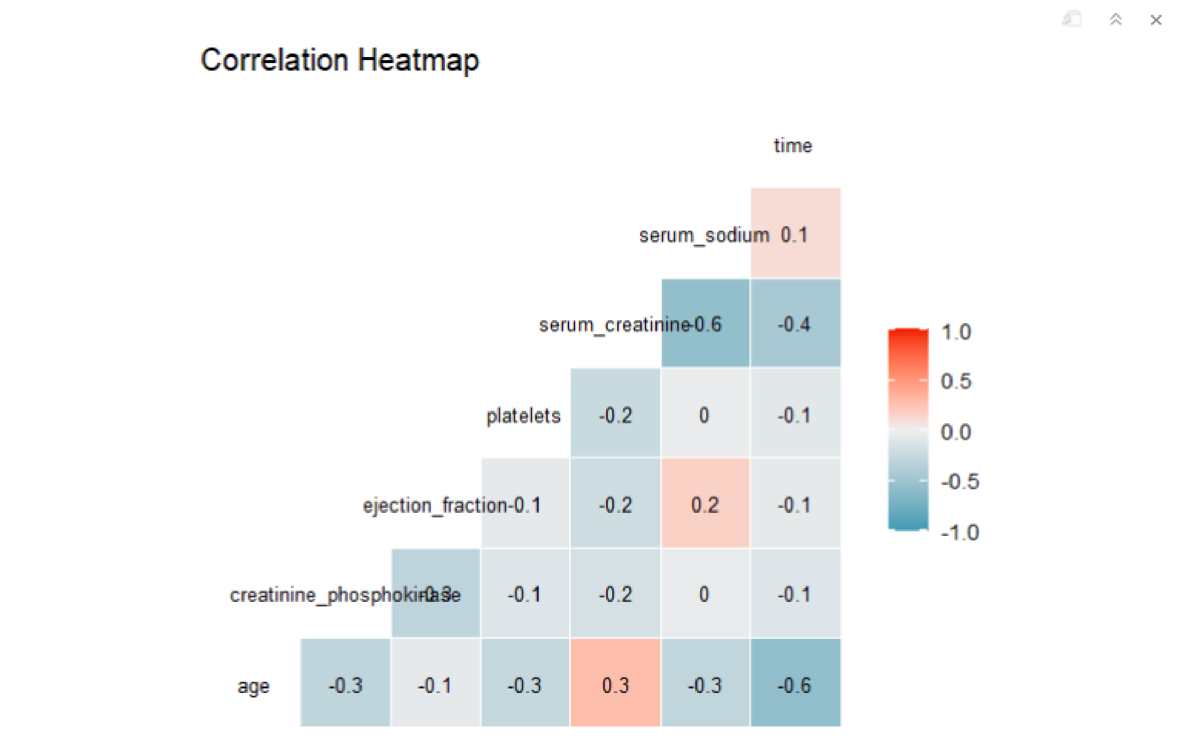
```
{r}
cor_matrix
```

```
                                age creatinine_phosphokinase ejection_fraction
platelets serum_creatinine serum_sodium         time
age                       1.00000000              -0.081583900        0.06009836
-0.05235437       0.15918713  -0.04596584 -0.224068420
creatinine_phosphokinase -0.08158390               1.000000000       -0.04407955
0.02446339      -0.01640848   0.05955016 -0.009345653
ejection_fraction         0.06009836              -0.044079554        1.00000000
0.07217747      -0.01130247   0.17590228  0.041729235
platelets                -0.05235437               0.024463389        0.07217747
1.00000000      -0.04119808   0.06212462  0.010513909
serum_creatinine          0.15918713              -0.016408480       -0.01130247
-0.04119808       1.00000000  -0.18909521 -0.149315418
serum_sodium             -0.04596584               0.059550156        0.17590228
0.06212462      -0.18909521   1.00000000  0.087640000
time                     -0.22406842              -0.009345653        0.04172924
0.01051391      -0.14931542   0.08764000  1.000000000
```

```{r}
ggcorr(cor_matrix, label = TRUE, label_size = 3,
       palette = "RdBu", geom = "tile", size = 3) +
  ggtitle("Correlation Heatmap")
```



```{r}
ggcorr(cor_matrix, label = TRUE, label_size = 3,
       palette = "RdBu", geom = "tile", size = 3) +
  ggtitle("Correlation Heatmap")
```

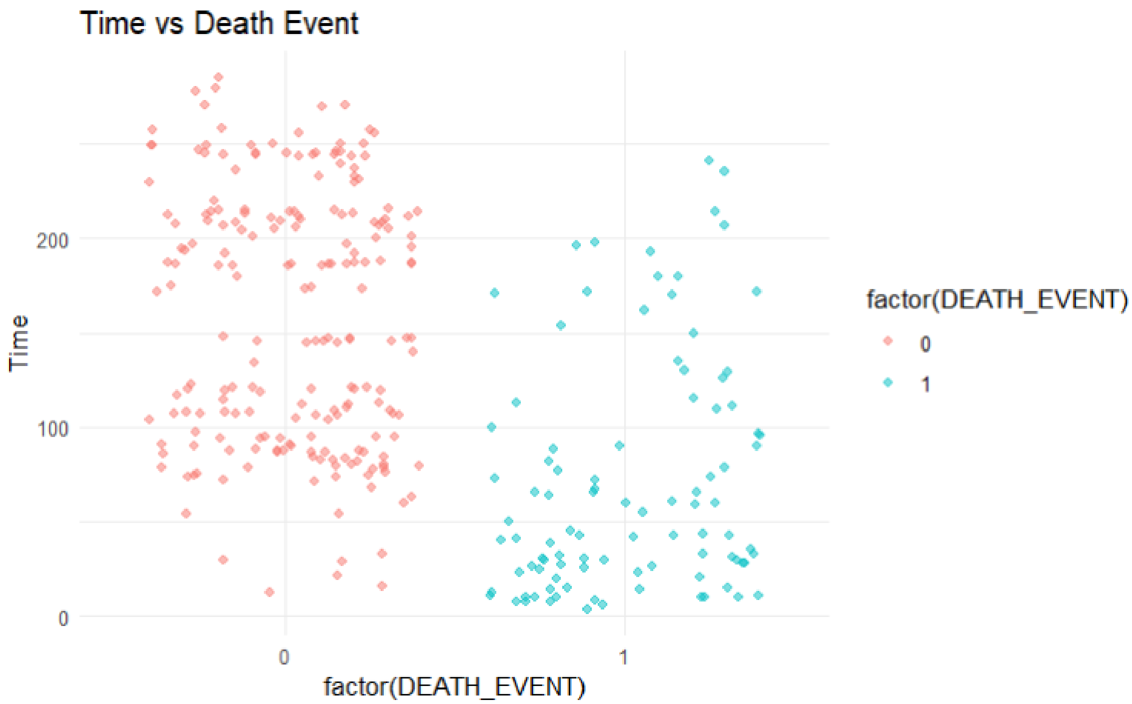


The correlation matrix shows largely weak relationships between the numerical variables, with the largest positive connection between ejection fraction and serum sodium (0.18) and the strongest negative correlation between age and time (-0.22). Serum creatinine and serum sodium had a moderate negative association (-0.19), which could indicate the effect of kidney disease on fluid and electrolyte balance in heart failure patients.

Scatter plots

To explore the potential relationship between continuous variables and the binary outcome (death event), creating scatter plots with the continuous variable on the x-axis and the death event (0 or 1) on the y-axis. This can provide a visual representation of how the values of the continuous variable may differ between those who died and those who survived.

```
{r}

ggplot(heart_failure_data, aes(x = factor(DEATH_EVENT), y = age, color = factor
(DEATH_EVENT))) +
  geom_jitter(alpha = 0.5) +
  labs(title = "Age vs Death Event", y = "Age") +
  theme_minimal()

ggplot(heart_failure_data, aes(x = factor(DEATH_EVENT), y =
creatinine_phosphokinase, color = factor(DEATH_EVENT))) +
  geom_jitter(alpha = 0.5) +
  labs(title = "Creatinine Phosphokinase vs Death Event", y = "Creatinine
Phosphokinase") +
  theme_minimal()

ggplot(heart_failure_data, aes(x = factor(DEATH_EVENT), y = ejection_fraction,
color = factor(DEATH_EVENT))) +
  geom_jitter(alpha = 0.5) +
  labs(title = "Ejection Fraction vs Death Event", y = "Ejection Fraction") +
  theme_minimal()

ggplot(heart_failure_data, aes(x = factor(DEATH_EVENT), y = platelets, color =
factor(DEATH_EVENT))) +
  geom_jitter(alpha = 0.5) +
  labs(title = "Platelets vs Death Event", y = "Platelets") +
  theme_minimal()

ggplot(heart_failure_data, aes(x = factor(DEATH_EVENT), y = serum_creatinine, color
= factor(DEATH_EVENT))) +
  geom_jitter(alpha = 0.5) +
  labs(title = "Serum Creatinine vs Death Event", y = "Serum Creatinine") +
  theme_minimal()

ggplot(heart_failure_data, aes(x = factor(DEATH_EVENT), y = serum_sodium, color =
```



No significant difference has been found in the features with respect to the death event.

When comparing time distribution with respect to death event, the death patients has tendency to have time below 0 and survival patients has higher time when compared.

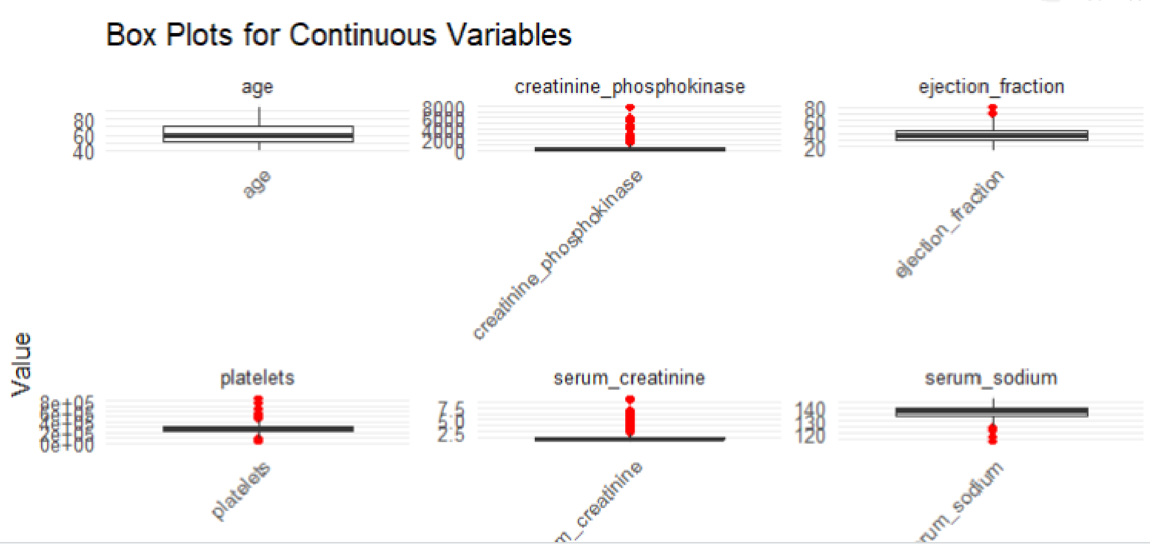
Check for Outliers

```
{r}
heart_failure_data_long <- heart_failure_data %>%
  pivot_longer(
    cols = c("age", "creatinine_phosphokinase", "ejection_fraction", "platelets",
"serum_creatinine", "serum_sodium", "time"),
    names_to = "variable",
```

Check for Outliers

```
{r}
heart_failure_data_long <- heart_failure_data %>%
  pivot_longer(
    cols = c("age", "creatinine_phosphokinase", "ejection_fraction", "platelets",
"serum_creatinine", "serum_sodium", "time"),
    names_to = "variable",
    values_to = "value"
  )
```

```
{r}
# Creating box plots for each variable using ggplot2
ggplot(heart_failure_data_long, aes(x = variable, y = value)) +
  geom_boxplot(outlier.color = "red") +
  facet_wrap(~ variable, scales = "free") +
  theme_minimal() +
  labs(title = "Box Plots for Continuous Variables", x = "", y = "Value") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Age: No outliers in the age column.

Creatinine Phosphokinase (CPK) Levels: The boxplot shows several very high CPK level values, which are considered outliers.

Ejection Fraction: Two patients have unusually high ejection fraction values compared to the rest, marked as outliers.

Platelet Count: Several patients have extremely high platelet counts, which are identified as outliers.

Serum Creatinine: The boxplot reveals several patients with very high serum creatinine levels, considered as outliers.

Serum Sodium: A few patients have unusually low serum sodium levels, marked as outliers.

Follow-up Period (time): No outliers in the follow-up period column.

Using standardized residuals to find outliers

```
{r}
library(glm2)
```

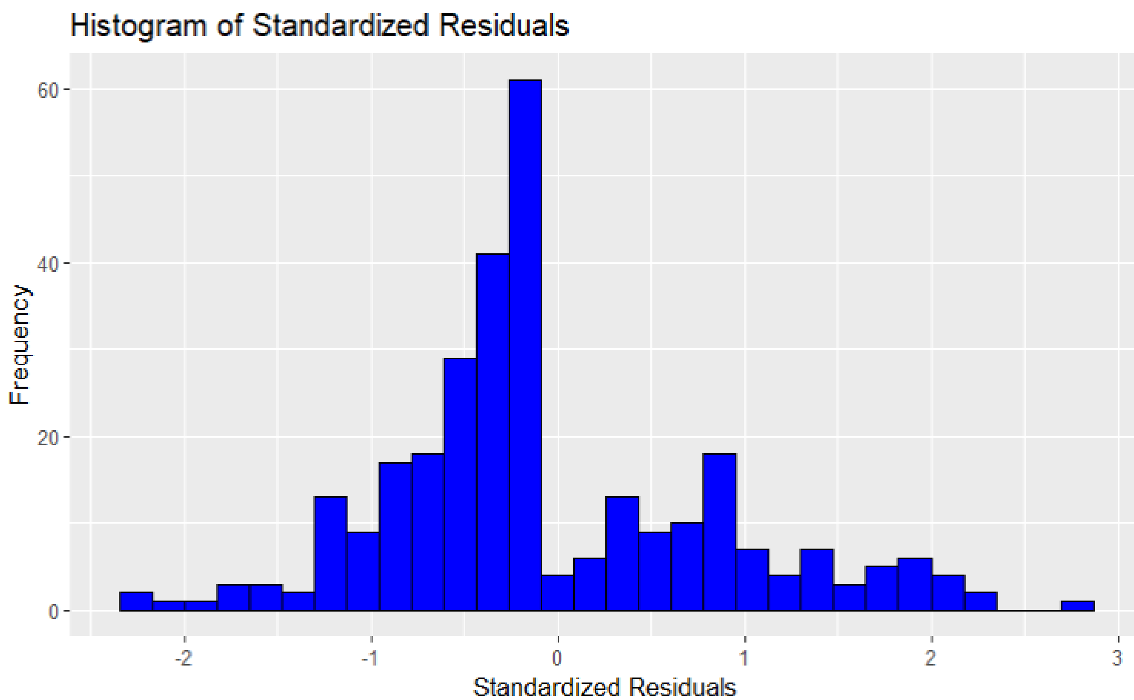
```
{r}
# Creating a logistic regression model
model <- glm(DEATH_EVENT ~ age + creatinine_phosphokinase + ejection_fraction +
platelets + serum_creatinine + serum_sodium + time,
             data = heart_failure_data, family = binomial())
```

```
{r}
#summary of the model
summary(model)
```

```
Call:
glm(formula = DEATH_EVENT ~ age + creatinine_phosphokinase +
```

```
{r}
# standardized residuals
heart_failure_data$std_residuals <- rstandard(model)
```

```
{r}
#plotting the standardized residuals
ggplot(heart_failure_data, aes(x = std_residuals)) +
  geom_histogram(bins = 30, fill = "blue", color = "black") +
  ggtitle("Histogram of Standardized Residuals") +
  xlab("Standardized Residuals") +
  ylab("Frequency")
```



Now, lets handle the outliers accordingly.

```
{r}
outliers <- heart_failure_data %>%
  filter(abs(x = std_residuals) > 1.96)
```

```
{r}
outliers
```

Description: df [11 × 14]

age <dbl>	anaemia <fctr>	creatinine_ph... <int>	diabetes <fctr>	ejection_fracti... <int>
65	1	52	0	25
60	0	2656	1	30
70	1	143	0	60
60	1	1082	1	45
50	1	2334	1	35
50	0	582	0	50
77	1	418	0	45
48	1	131	1	30
55	0	2017	0	25
65	1	258	1	25

1-10 of 11 rows | 1-5 of 14 columns   Previous 1 2 Next

The model underpredicts survival probability for patients with very high CPK levels (e.g., observations 2 and 5) and severe kidney dysfunction (observation 4), while overpredicting survival for patients with low ejection fractions (observations 9 and 11) and electrolyte imbalances such as hyponatremia (observations 8 and 10). Furthermore, the model had difficulty with situations involving older patients (e.g., observations 3 and 7) and those with various comorbidities such as anemia, diabetes, and high blood pressure.

Lets remove outliers completely

Lets remove outliers completely.

```
{r}
#filtering
heart_failure_data_cleaned <- heart_failure_data %>%
  filter(abs(std_residuals)<= 1.96)
```

```
{r}
head(heart_failure_data_cleaned)
```

Description: df [6 × 14]

	age <dbl>	anaemia <fctr>	creatinine_... <int>	diabetes <fctr>	ejection_fr... <int>
1	75	0	582	0	20
2	55	0	7861	0	38
3	65	0	146	0	20
4	50	1	111	0	20
5	65	1	160	1	20
6	90	1	47	0	40

6 rows | 1-6 of 14 columns

```
{r}
str(heart_failure_data_cleaned)
```

```
'data.frame':	288 obs. of  14 variables:
 $ age                     : num  75 55 65 50 65 90 75 60 65 80 ...
 $ anaemia                 : Factor w/ 2 levels "0","1": 1 1 1 2 2 2 2 2 1 2 ...
 $ creatinine_phosphokinase: int  582 7861 146 111 160 47 246 315 157 123 ...
 $ diabetes                : Factor w/ 2 levels "0","1": 1 1 1 1 2 1 1 2 1 1 ...
 $ ejection_fraction       : int  20 38 20 20 20 40 15 60 65 35 ...
 $ high_blood_pressure     : Factor w/ 2 levels "0","1": 2 1 1 1 1 2 1 1 1 2 ...
 $ platelets               : num  265000 263358 162000 210000 327000 ...
 $ serum_creatinine        : num  1.9 1.1 1.3 1.9 2.7 2.1 1.2 1.1 1.5 9.4 ...
 $ serum_sodium            : int  130 136 129 137 116 132 137 131 138 133 ...
 $ sex                     : Factor w/ 2 levels "0","1": 2 2 2 2 1 2 2 2 1 2 ...
 $ smoking                 : Factor w/ 2 levels "0","1": 1 1 2 1 1 2 1 2 1 2 ...
```

```r
{r}
cleaned_model <- glm(DEATH_EVENT ~ age + creatinine_phosphokinase +
ejection_fraction + platelets + serum_creatinine + serum_sodium + time,
                     data = heart_failure_data_cleaned, family = binomial())
```

```r
{r}
summary(cleaned_model)
```

```
Call:
glm(formula = DEATH_EVENT ~ age + creatinine_phosphokinase + 
    ejection_fraction + platelets + serum_creatinine + serum_sodium + 
    time, family = binomial(), data = heart_failure_data_cleaned)

Coefficients:
                           Estimate Std. Error z value Pr(>|z|)    
(Intercept)               1.191e+01  6.470e+00   1.842  0.06553 .  
age                       5.640e-02  1.856e-02   3.039  0.00237 ** 
creatinine_phosphokinase  2.087e-04  2.009e-04   1.039  0.29887    
ejection_fraction        -9.492e-02  1.985e-02  -4.782 1.74e-06 ***
platelets                 8.317e-08  2.100e-06   0.040  0.96840    
serum_creatinine          1.098e+00  2.275e-01   4.825 1.40e-06 ***
serum_sodium             -8.159e-02  4.560e-02  -1.789  0.07359 .  
time                     -3.228e-02  4.527e-03  -7.130 1.00e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 354.53  on 287  degrees of freedom
Residual deviance: 159.51  on 280  degrees of freedom
AIC: 175.51

Number of Fisher Scoring iterations: 6
```

Standardization

Standardization

```
{r}

# Standardizing continuous variables using Scale
heart_failure_data_cleaned$age <- scale(heart_failure_data_cleaned$age)

heart_failure_data_cleaned$creatinine_phosphokinase <- scale
(heart_failure_data_cleaned$creatinine_phosphokinase)

heart_failure_data_cleaned$ejection_fraction <- scale
(heart_failure_data_cleaned$ejection_fraction)

heart_failure_data_cleaned$platelets <- scale(heart_failure_data_cleaned$platelets)

heart_failure_data_cleaned$serum_creatinine <- scale
(heart_failure_data_cleaned$serum_creatinine)

heart_failure_data_cleaned$serum_sodium <- scale
(heart_failure_data_cleaned$serum_sodium)

heart_failure_data_cleaned$time <- scale(heart_failure_data_cleaned$time)
```

```
{r}
# Summary to check mean and standard deviation
summary(heart_failure_data_cleaned[c("age", "creatinine_phosphokinase",
"ejection_fraction", "platelets", "serum_creatinine", "serum_sodium", "time")])
```

```
        age.V1          creatinine_phosphokinase.V1 ejection_fraction.V1
 platelets.V1       serum_creatinine.V1    serum_sodium.V1
 Min.   :-1.7405685   Min.   :-0.560557           Min.   :-2.047564    Min.
:-2.427675    Min.   :-0.865644   Min.   :-5.378043
 1st Qu.:-0.8237387   1st Qu.:-0.465637           1st Qu.:-0.692909    1st
Qu.:-0.522450    1st Qu.:-0.470767   1st Qu.:-0.603875
 Median :-0.0736051   Median :-0.329447           Median :-0.015581    Median
:-0.017334    Median :-0.273329   Median : 0.078149
 Mean   : 0.0000000   Mean   : 0.000000           Mean   : 0.000000    Mean   :
0.000000    Mean   : 0.000000   Mean   : 0.000000
 3rd Qu.: 0.7598766   3rd Qu.: 0.016185           3rd Qu.: 0.577081    3rd Qu.:
0.396404    3rd Qu.: 0.022829   3rd Qu.: 0.760172
```

The mean of each variable is 0. S.D of each variable is 1. So, standardization succesfull.

Statistical tests, Assumptions, and Alternate Tests

1. Chi-Square Test for Categorical Variables

Many existing researches that High blood pressure, anemia are major factors for heart diseases. So, lets check the relation between high blood pressure, anemia and death.

Fuchs FD, Whelton PK. High blood pressure and cardiovascular disease. *Hypertension*. 2020;75(2):285-292. doi:10.1161/hypertensionaha.119.14240

The role of anemia in the progression of congestive heart failure. Is there a place for erythropoietin and intravenous iron? PubMed. Published December 1, 2004. https://pubmed.ncbi.nlm.nih.gov/15593047/

```
{r}
# Chi-squared Test for Anemia and Death Event
table_anemia <- table(heart_failure_data_cleaned$anaemia,
heart_failure_data_cleaned$DEATH_EVENT)
chisq.test(table_anemia)
```

```
        Pearson's Chi-squared test with Yates' continuity correction

 data:  table_anemia
 X-squared = 0.69584, df = 1, p-value = 0.4042
```

```
{r}
# Chi-squared Test for High Blood Pressure and Death Event
table_hbp <- table(heart_failure_data_cleaned$high_blood_pressure,
heart_failure_data_cleaned$DEATH_EVENT)
chisq.test(table_hbp)
```

```
{r}
# Chi-squared Test for diabetes and Death Event
table_dbts <- table(heart_failure_data_cleaned$diabetes,
heart_failure_data_cleaned$DEATH_EVENT)
chisq.test(table_dbts)
```

```
        Pearson's Chi-squared test with Yates' continuity correction

 data:  table_dbts
 X-squared = 2.3604e-30, df = 1, p-value = 1
```

```
{r}
# Fisher's exact test for Sex and DEATH_EVENT
table_sex <- table(heart_failure_data_cleaned$sex,
heart_failure_data_cleaned$DEATH_EVENT)
fisher.test(table_sex)
```

```

	Fisher's Exact Test for Count Data

data:  table_sex
p-value = 1
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.5962372 1.8377372
sample estimates:
odds ratio
  1.040889
```

```
{r}
# Fisher's exact test for Smoking and DEATH_EVENT
table_smoking <- table(heart_failure_data_cleaned$smoking,
heart_failure_data_cleaned$DEATH_EVENT)
fisher.test(table_smoking)
```

```

	Fisher's Exact Test for Count Data

data:  table_smoking
p-value = 1
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.5615575 1.7523316
sample estimates:
odds ratio
 0.9979527
```

The tests did not find a statistically significant link between gender (p-value = 1) and smoking status (p-value = 1) and the death event in this dataset of heart failure patients.

The test for sex has a p-value of 1, indicating a complete lack of association between gender and mortality in this particular group of heart failure patients.

Similarly, the test for smoking status also has a p-value of 1, suggesting no significant relationship between smoking status and the death event in this dataset.

The p-value of 1 for both variables implies that the observed data is entirely consistent with the null hypothesis of no association between these variables and the death event.

The 95% confidence intervals for the odds ratios include 1 for both sex (0.5962372, 1.8377372) and smoking (0.5615575, 1.7523316), supporting the conclusion that these variables have no statistically significant association with mortality in this group of heart failure patients.

3. Logistic Regression Model Test for Continous Variables

```
{r}
continous_variables <- c("age", "ejection_fraction", "serum_sodium",
"creatinine_phosphokinase", "platelets", "serum_creatinine", "time")
```

```
{r}
significance_level <- 0.05
```

```
{r}
# Performing logistic regression model tests
for (variable in continous_variables) {
  cat("\n*****************************START OF TEST***********************\n\n")
  print(paste("Step 1: Defining Hypotheses for", variable))
  print(paste("Null Hypothesis: There is no significant association between",
variable, "and DEATH_EVENT."))
  print(paste("Alternative Hypothesis: There is a significant association between",
variable, "and DEATH_EVENT."))
  cat("\n----------------------------------\n\n")
  print("Step 2: Performing the test...")
```

```
******************************START OF TEST************************

[1] "Step 1: Defining Hypotheses for age"
[1] "Null Hypothesis: There is no significant association between age and
DEATH_EVENT."
[1] "Alternative Hypothesis: There is a significant association between age and
DEATH_EVENT."

----------------------------------

[1] "Step 2: Performing the test..."

----------------------------------

[1] "Step 3: Analyzing the test results..."

Call:
glm(formula = DEATH_EVENT ~ ., family = binomial(link = "logit"),
    data = heart_failure_data_cleaned[, c("DEATH_EVENT", variable)])

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.8851     0.1356  -6.526 6.77e-11 ***
age           0.6107     0.1374   4.443 8.86e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 354.53  on 287  degrees of freedom
Residual deviance: 333.00  on 286  degrees of freedom
AIC: 337

Number of Fisher Scoring iterations: 4


----------------------------------

[1] "Step 4: Making a conclusion..."
[1] "We reject the null hypothesis. There is a significant association."

******************************END OF TEST************************
```

```
*****************************START OF TEST***********************

[1] "Step 1: Defining Hypotheses for ejection_fraction"
[1] "Null Hypothesis: There is no significant association between
ejection_fraction and DEATH_EVENT."
[1] "Alternative Hypothesis: There is a significant association between
ejection_fraction and DEATH_EVENT."

----------------------------------

[1] "Step 2: Performing the test..."

----------------------------------

[1] "Step 3: Analyzing the test results..."

Call:
glm(formula = DEATH_EVENT ~ ., family = binomial(link = "logit"),
    data = heart_failure_data_cleaned[, c("DEATH_EVENT", variable)])

Coefficients:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)        -0.9189     0.1396  -6.581 4.66e-11 ***
ejection_fraction  -0.7150     0.1573  -4.545 5.49e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 354.53  on 287  degrees of freedom
Residual deviance: 329.88  on 286  degrees of freedom
AIC: 333.88

Number of Fisher Scoring iterations: 4


----------------------------------

[1] "Step 4: Making a conclusion..."
[1] "We reject the null hypothesis. There is a significant association."

*****************************END OF TEST***********************
```

```
*****************************START OF TEST***********************

[1] "Step 1: Defining Hypotheses for serum_sodium"
[1] "Null Hypothesis: There is no significant association between serum_sodium
and DEATH_EVENT."
[1] "Alternative Hypothesis: There is a significant association between
serum_sodium and DEATH_EVENT."

----------------------------------

[1] "Step 2: Performing the test..."

----------------------------------

[1] "Step 3: Analyzing the test results..."

Call:
glm(formula = DEATH_EVENT ~ ., family = binomial(link = "logit"),
    data = heart_failure_data_cleaned[, c("DEATH_EVENT", variable)])

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.8517     0.1317  -6.469  9.9e-11 ***
serum_sodium  -0.4484     0.1366  -3.282  0.00103 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 354.53  on 287  degrees of freedom
Residual deviance: 342.69  on 286  degrees of freedom
AIC: 346.69

Number of Fisher Scoring iterations: 4


----------------------------------

[1] "Step 4: Making a conclusion..."
[1] "We reject the null hypothesis. There is a significant association."

*****************************END OF TEST***********************
```

```
*****************************START OF TEST***********************

[1] "Step 1: Defining Hypotheses for creatinine_phosphokinase"
[1] "Null Hypothesis: There is no significant association between
creatinine_phosphokinase and DEATH_EVENT."
[1] "Alternative Hypothesis: There is a significant association between
creatinine_phosphokinase and DEATH_EVENT."

----------------------------------

[1] "Step 2: Performing the test..."

----------------------------------

[1] "Step 3: Analyzing the test results..."

Call:
glm(formula = DEATH_EVENT ~ ., family = binomial(link = "logit"),
    data = heart_failure_data_cleaned[, c("DEATH_EVENT", variable)])

Coefficients:
                         Estimate Std. Error z value Pr(>|z|)
(Intercept)               -0.8232     0.1282  -6.421 1.35e-10 ***
creatinine_phosphokinase   0.1166     0.1208   0.965    0.334
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 354.53  on 287  degrees of freedom
Residual deviance: 353.62  on 286  degrees of freedom
AIC: 357.62

Number of Fisher Scoring iterations: 4


----------------------------------

[1] "Step 4: Making a conclusion..."
[1] "We fail to reject the null hypothesis. There is no significant association."

*****************************END OF TEST***********************
```

```
[1] "Step 2: Performing the test..."

---------------------------------

[1] "Step 3: Analyzing the test results..."

Call:
glm(formula = DEATH_EVENT ~ ., family = binomial(link = "logit"),
    data = heart_failure_data_cleaned[, c("DEATH_EVENT", variable)])

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.8222     0.1281  -6.420 1.36e-10 ***
platelets    -0.0785     0.1316  -0.596    0.551
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 354.53  on 287  degrees of freedom
Residual deviance: 354.16  on 286  degrees of freedom
AIC: 358.16

Number of Fisher Scoring iterations: 4


---------------------------------

[1] "Step 4: Making a conclusion..."
[1] "We fail to reject the null hypothesis. There is no significant association."

*****************************END OF TEST***********************


*****************************START OF TEST***********************

[1] "Step 1: Defining Hypotheses for serum_creatinine"
[1] "Null Hypothesis: There is no significant association between
serum_creatinine and DEATH_EVENT."
[1] "Alternative Hypothesis: There is a significant association between
serum_creatinine and DEATH_EVENT."

---------------------------------
```

```
[1] We reject the null hypothesis. There is a significant association.

*****************************END OF TEST***********************
```

Feature Name	P-value	Finding
age	< 0.001	Significant association
ejection_fraction	< 0.001	Significant association
serum_sodium	0.001	Significant association
creatinine_phosphokinase	0.334	No significant association
platelets	0.551	No significant association
serum_creatinine	< 0.001	Significant association
time	< 0.001	Significant association

Logistic Model for Predictive Analysis

Split the data into test and train using 75% Train

```
{r}
set.seed(32)  # seed number for reproducibility
```

```
{r}
training_indices <- sample(1:nrow(heart_failure_data_cleaned), 0.75 * nrow
(heart_failure_data_cleaned))
```

```
{r}
train_data <- heart_failure_data_cleaned[training_indices, ]
test_data <- heart_failure_data_cleaned[-training_indices, ]
```

```
  formula <- as.formula(paste("DEATH_EVENT ~", variables[i]))
  model <- glm(formula, family = binomial(link = "logit"), data = train_data)
  y_pred <- ifelse(predict(model, type = "response") > 0.5, 1, 0)
  accuracies[i] <- mean(y_pred == train_data$DEATH_EVENT)
  print(paste("Accuracy for", variables[i], ":", accuracies[i]))
}
```

```
[1] "Accuracy for age : 0.712962962962963"
[1] "Accuracy for ejection_fraction : 0.740740740740741"
[1] "Accuracy for serum_creatinine : 0.712962962962963"
[1] "Accuracy for serum_sodium : 0.689814814814815"
[1] "Accuracy for time : 0.851851851851852"
```

```
{r}
results <- data.frame(Variable = variables, Accuracy = accuracies)


knitr::kable(results, caption = "Logistic Regression Model Accuracies for Each
Variable")
```

With Multiple Variables

Fit the model using train data

Considering only the elements which we found that are significantly different.

```{r}
# Fit logistic regression model on training data with only our finding columns
model <- glm(DEATH_EVENT ~ age  + ejection_fraction + serum_creatinine +
serum_sodium + time,
             family = binomial(link = "logit"), data = train_data)
```

```{r}
# Summary of the model
summary(model)
```

```
Call:
glm(formula = DEATH_EVENT ~ age + ejection_fraction + serum_creatinine +
    serum_sodium + time, family = binomial(link = "logit"), data = train_data)

Coefficients:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)        -1.9484     0.3257  -5.982 2.20e-09 ***
age                 0.5216     0.2406   2.168  0.03019 *
ejection_fraction  -1.2453     0.2760  -4.512 6.42e-06 ***
serum_creatinine    0.8945     0.2867   3.120  0.00181 **
serum_sodium       -0.4392     0.2227  -1.972  0.04856 *
time               -2.3273     0.3709  -6.275 3.49e-10 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 270.63  on 215  degrees of freedom
Residual deviance: 130.03  on 210  degrees of freedom
AIC: 142.03

Number of Fisher Scoring iterations: 6
```

Test the model performance on test data

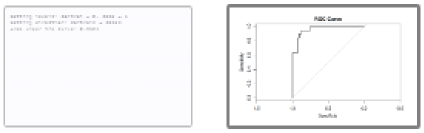
```
{r}
# Predicting on test data
test_data$predicted_prob <- predict(model, newdata = test_data, type = "response")
test_data$predicted_class <- ifelse(test_data$predicted_prob > 0.5, 1, 0)
```

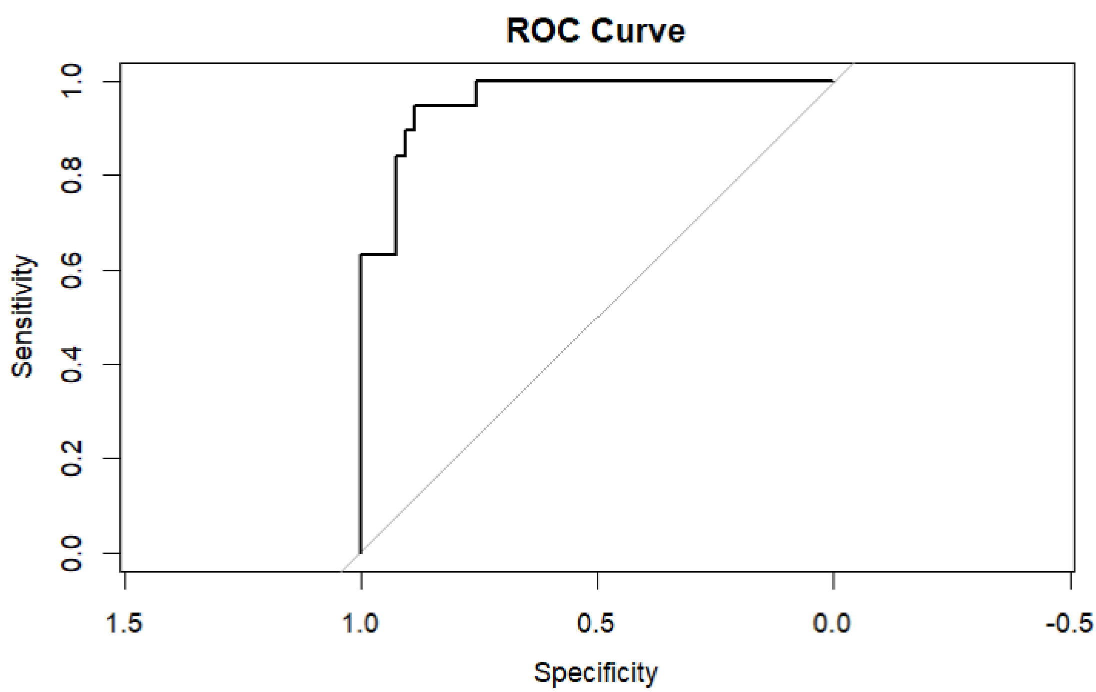
Confusion Matrix

```
{r}
# Confusion matrix

confusionMatrix(factor(test_data$predicted_class), factor(test_data$DEATH_EVENT))
```

```
Confusion Matrix and Statistics

          Reference
Prediction  0  1
         0 52  7
         1  1 12

               Accuracy : 0.8889
                 95% CI : (0.7928, 0.9508)
    No Information Rate : 0.7361
    P-Value [Acc > NIR] : 0.001268

                  Kappa : 0.6818

 Mcnemar's Test P-Value : 0.077100

            Sensitivity : 0.9811
            Specificity : 0.6316
         Pos Pred Value : 0.8814
         Neg Pred Value : 0.9231
             Prevalence : 0.7361
         Detection Rate : 0.7222
   Detection Prevalence : 0.8194
      Balanced Accuracy : 0.8064

       'Positive' Class : 0
```

```{r}
# ROC Curve
roc_response <- roc(test_data$DEATH_EVENT, test_data$predicted_prob)
plot(roc_response, main="ROC Curve")
auc(roc_response)
```





Conclusions on Model Performance